

Linknet Corporation, Enterprise, Fiber, Media

Network Topology and Infrastructure Architecture

Enterprise infrastructure documentation prepared from actual project source code, deployment references, runtime configuration files, and service integration evidence.

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Prepared By	Media Indo Trend
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Company / Organization	LinkNet Corporation, Enterprise, Fiber, Media / PT Link Net Tbk
Document Purpose	Audit, security review, deployment review, handover, and internal IT documentation.

Evidence Handling Note

This document intentionally avoids unsupported assumptions. Items not evidenced in the repository are marked as Not Found in Current Configuration or Deployment Detail Not Available.

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1. Executive Summary

Infrastructure Assessment Summary

The project is implemented as a multi-application workspace: public Next.js website, Next.js admin CMS, Express.js backend API, PostgreSQL via Prisma, and a standalone S3-oriented filemanager service. Dockerfiles exist for all deployable components. Backend Kubernetes references exist, but full production network, DNS, load balancer, firewall, and database HA details are not fully available in the current configuration.

This document was generated from actual implementation evidence found in source code, Dockerfiles, Kubernetes manifests, CI/CD workflow files, package manifests, Prisma schema, and environment configuration keys. Where deployment infrastructure is not present, the document explicitly marks the gap instead of inventing an architecture.

Scope Area	Finding
Architecture Type	Multi-application / service-separated architecture within one repository: public web, admin frontend, backend API, and filemanager service.
Primary Runtime	Node.js applications using Next.js and Express.js. Backend API exposes /api/v1 and health endpoints.
Persistence	PostgreSQL datasource via Prisma. Database server host/port/HA detail not disclosed in current configuration.
Storage	Local upload path plus AWS S3 and Azure Blob integration paths. Filemanager has concrete AWS S3 bucket/CDN configuration.
Cloud Services	AWS S3, AWS CloudFront, AWS ECR indication, Azure Key Vault CSI template, optional Azure Blob, GA4 Data API and external LinkNet upstream APIs.
Containerization	Dockerfiles found for backend, frontend, web, and filemanager. Backend Kubernetes manifests found. docker-compose not found.
Key Gaps	DNS provider, public IP, production load balancer, firewall/WAF, network policy, S3 IAM policy, database HA/topology, and frontend ingress details are not found.

2. Document Information

Field	Value
Project Name	Revamp LinkNet Corporation, Enterprise, Fiber, Media
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Field	Value
Classification	Internal IT Documentation
Source Basis	Current source code and configuration under Azure Repository / Azure Devops Service
Review Method	Static analysis of repository files, environment keys, deployment descriptors, middleware, routes, service integrations, package manifests, and Prisma schema.

3. Infrastructure Overview

The implementation separates public website rendering, administrative CMS access, API/business logic, file operations, persistence, and cloud/external integrations. Backend manifests indicate containerized deployment and horizontal scaling capability for the API tier, while frontend and filemanager production orchestration details are not fully present in the repository.

3.1 High-Level Infrastructure Summary

Area	Status	Implementation Evidence
Frontend Server	Found	Next.js public web under web/ and admin CMS under frontend/; Docker standalone mode; ports 3000 and 3001 respectively.
Backend Server	Found	Express.js/TypeScript API under backend/; API prefix /api/v1; health endpoints /health and /ready; Docker exposes port 5000.
Database Server	Partial	PostgreSQL via Prisma DATABASE_URL. Concrete DB host, subnet, HA, backup, and network ACL detail not found.
Reverse Proxy / Load Balancer	Partial	backend/k8s/ingress.yaml uses nginx ingress and TLS annotations. Production edge/LB for frontend/public web not found.
Storage Service	Found	AWS S3 service and Azure Blob optional service found. Filemanager production env references AWS S3 bucket and CloudFront CDN.
Cloud Services	Found	AWS S3/CloudFront/ECR indications, Azure Key Vault CSI template, Azure Blob optional, GA4 Data API.
Container / Orchestration	Partial	Dockerfiles for all components; backend Kubernetes Deployment/Service/HPA/Ingress/KeyVault CSI templates; no docker-compose.
Queue / Broker	Partial	Bull/ioredis dependencies and REDIS_ENABLED feature flag found. Runtime Redis deployment detail not available.

3.2 Application Deployment Overview

- Public Web: Next.js application built with standalone output and intended to call backend API through NEXT_PUBLIC_API_URL or API_INTERNAL_URL.

- Admin CMS: Next.js application built with standalone output, auth-enabled flag, and API base URL configured through environment variables.
- Backend API: Express.js application with middleware chain for security headers, CORS, cookies, CSRF, rate limiting, auth, RBAC, compression, audit logging, and route modules.
- Filemanager: standalone Express service for S3 upload, listing, delete, and signed URL functions; production API key enforcement fails closed when API_KEY is missing.
- Backend Kubernetes: replicas 3, ClusterIP service, HPA min 3 max 10, liveness/readiness probes, securityContext, ConfigMap/Secret references, and nginx ingress TLS template.

3.3 Environment Overview

Environment Indicator	Evidence	Assessment
Development	Local .env files and package scripts found across components.	Development environment exists by convention, but local network topology is not fully documented.
Staging	No explicit staging descriptor found.	Not Found in Current Configuration.
Production	.env.production files, Dockerfiles, CI workflows, backend Kubernetes manifests, and production domains found.	Production intent is evidenced, but edge/network/cloud resource configuration is incomplete.
CI/CD	GitHub Actions build workflow, Trivy scan workflow, and web Azure Pipeline/ECR pipeline files found.	Build/security scanning is evidenced; final deployment promotion workflow is partial.

4. Technology and Infrastructure Stack

Component	Package Name	Version	Dependencies	Key Packages
Backend API	linknet-backend	1.0.0	51 runtime / 27 dev	express@^4.21.3, @prisma/client@^5.7.1, prisma@^5.22.0 (dev), typescript@^5.3.3 (dev), @aws- sdk/client-s3@^3.1001.0, @azure/storage-blob@^12.29.1, @azure/keyvault-secrets@^4.10.0, @google-analytics/data@^5.2.1, bull@^4.16.5, ioredis@^5.3.2
Admin Frontend	free-nextjs-admin-dashboard	2.2.2	45 runtime / 11 dev	next@16.2.4, react@^19.2.0, typescript@^5.9.3 (dev)
Public Web	linknet_web	0.1.0	12 runtime / 12 dev	next@16.2.4, react@19.2.3, typescript@^5.9.3 (dev)
Standalone File Manager	filemanager	1.0.0	10 runtime / 11 dev	express@^4.19.2, typescript@^5.4.5 (dev), @aws- sdk/client-s3@^3.600.0

Infrastructure Category	Implementation Found	Detail
Frontend Server	Next.js	Public web under web/ exposes 3000; admin CMS under frontend/ exposes 3001; both use standalone Docker build pattern.
Backend Server	Node.js / Express.js	backend/src/server.ts initializes Express with security middleware and route modules under /api/v1.
Database	PostgreSQL via Prisma	backend/prisma/schema.prisma datasource provider postgresql and DATABASE_URL environment variable.
Reverse Proxy / LB	Partial	backend/k8s/ingress.yaml references nginx ingress and TLS; production LB/CDN provider for app edge not found.
Cloud Services	AWS and Azure references	AWS S3/CloudFront/ECR, Azure Key Vault CSI template, Azure Blob optional, GA4 Data API.
Storage	Local, S3, Azure Blob	Upload storage service selects local, azure, or s3; filemanager uses AWS S3 directly.
CDN	Partial	CloudFront CDN URL found for filemanager. CDN for primary site/app edge not found.
Queue / Broker	Optional Redis/Bull	Bull/ioredis code paths found; runtime Redis deployment not found.
Container / Orchestration	Docker + partial Kubernetes	Dockerfiles for all apps; backend Kubernetes Deployment/Service/HPA/Ingress; no docker-compose.

5. Network Topology Diagram

Diagram Interpretation

The diagram includes required topology elements where evidence exists. Public load balancer, DNS provider, firewall, WAF, network security group, database subnet, and frontend ingress details are marked as partial or unavailable because they are not defined in the uploaded configuration.

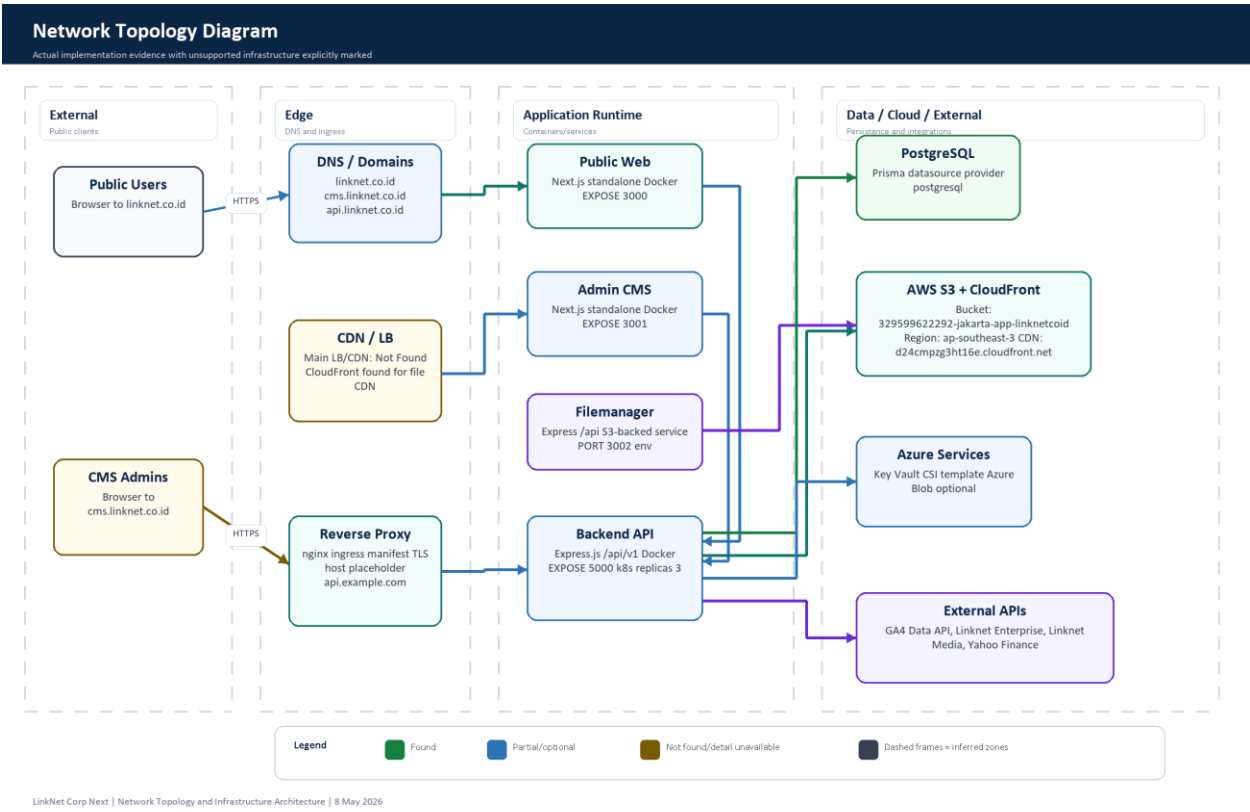


Figure 1. Network Topology Diagram based on current repository evidence

Topology Node	Evidence	Status / Notes
User / Client	Browser access to public site and CMS routes.	External public client boundary.
Internet / Public Access	Domains and HTTPS URLs in environment files.	Public access expected.
DNS / Domain	linknet.co.id, cms.linknet.co.id, api.linknet.co.id values found.	DNS provider and records Not Found in Current Configuration.
CDN / Load Balancer	CloudFront CDN URL found for filemanager; main edge LB not present.	Partial. Primary CDN/LB Deployment Detail Not Available.
Reverse Proxy	backend/k8s/ingress.yaml references nginx ingress and TLS annotations.	Backend only; frontend ingress Not Found in Current Configuration.
Frontend Applications	web/ and frontend/ Next.js Dockerfiles and env config.	Public web and admin CMS separated by app and port.
Backend API	backend/ Express API, Dockerfile and k8s deployment.	API server on port 5000, /api/v1 prefix.
Database Server	Prisma PostgreSQL datasource.	Host/subnet/HA/firewall detail Deployment Detail Not Available.
Storage Service	S3, Azure Blob and local upload implementations.	S3 bucket/CDN concrete in filemanager config; backend bucket env placeholders.
External Integrations	GA4, LinkNet upstream APIs, Yahoo Finance, optional email.	Outbound HTTPS integrations found; production egress controls Not Found.

Topology Node	Evidence	Status / Notes
Internal / Private Network	Kubernetes service/ClusterIP and service-to-database code boundaries.	Inferred from deployment config; concrete VPC/VNet/subnet not found.
Docker / Container Communication	Dockerfiles and backend Kubernetes service manifest.	docker-compose or full container network definition Not Found.

6. Server and Service Communication

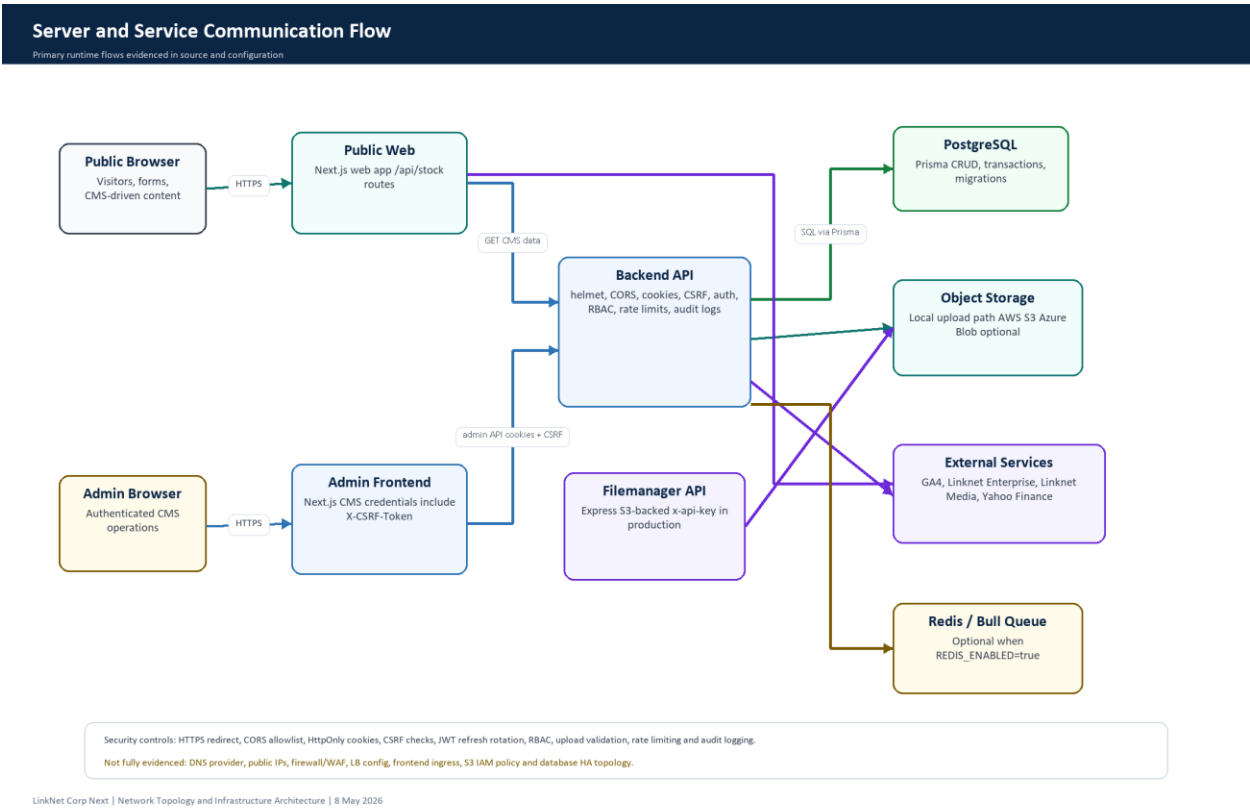


Figure 2. Server and Service Communication Flow

Flow	Implementation Evidence	Communication Notes
Frontend to Backend	frontend service layer and web CMS API clients use NEXT_PUBLIC_API_URL/API_INTERNAL_URL.	HTTPS expected through public domains. Admin sends credentials include, Authorization where available, and X-CSRF-Token for unsafe requests.
Backend to Database	Prisma client uses DATABASE_URL and PostgreSQL datasource.	SQL communication handled by Prisma. DB host, private subnet, SSL mode, and pooler detail not found.
Backend to Storage	storageService selects local/azure/s3; s3Service and azureStorage.service exist.	Uploads may go local disk, S3, or Azure Blob depending on STORAGE_DRIVER and environment variables.

Flow	Implementation Evidence	Communication Notes
Filemanager to Storage	filemanager/src services use @aws-sdk/client-s3 and CloudFront CDN URL.	S3 object operations under standalone /api service. Production API key required.
Backend to External APIs	Linknet Enterprise controller, Linknet Media controller, GA4 service.	Outbound HTTPS flows. Credential values are not disclosed; allowlist/egress firewall not found.
Web to Yahoo Finance	web/app/api/stock/quote and historical routes use yahoo-finance2.	Public web API route acts as proxy with validation/rate limiting.
Authentication Communication	JWT access/refresh, bcrypt, cookies, CSRF, MFA and RBAC middleware.	Internal auth service only; OAuth provider Not Found in Current Configuration.
Internal Service Communication	Backend Kubernetes Service 80 to container 5000; app-to-filemanager calls not clearly wired.	Service discovery and private DNS detail Deployment Detail Not Available.

7. Environment and Configuration

7.1 Environment Variables

Component	File	Environment Keys Found
Backend	backend/.env.production	API_PREFIX, API_VERSION, AWS_ACCESS_KEY_ID, AWS_REGION, AWS_S3_BUCKET, AWS_S3_ENDPOINT, AWS_S3_PUBLIC_URL, AWS_SECRET_ACCESS_KEY, AZURE_CLIENT_ID, AZURE_CLIENT_SECRET, AZURE_KEY_VAULT_URL, AZURE_STORAGE_ACCOUNT_NAME, AZURE_STORAGE_CONNECTION_STRING, AZURE_STORAGE_CONTAINER, AZURE_STORAGE_CONTAINER_NAME, AZURE_TENANT_ID, BASE_URL, CORS_CREDENTIALS, CORS_ORIGIN, DATABASE_URL, ENABLE_FILE_LOGS, FORCE_HTTPS, ... (38 total keys)
Admin Frontend	frontend/.env.production	NEXT_PUBLIC_API_URL, NEXT_PUBLIC_AUTH_ENABLED, NEXT_PUBLIC_SITE_URL
Admin Frontend Local	frontend/.env.local	NEXT_DISABLE_TYPE_CHECKING, NEXT_PRIVATE_BUILD_WORKER, NEXT_TELEMETRY_DISABLED
Public Web	web/.env.production	NEXT_PUBLIC_API_URL
Public Web Local	web/.env	NEXT_PUBLIC_API_URL
Filemanager	filemanager/.env.production	ALLOWED_ORIGINS, AWS_BUCKET_NAME, AWS_REGION, CDN_URL, MAX_FILE_SIZE_MB, NODE_ENV, PORT
Filemanager Local	filemanager/.env	ALLOWED_ORIGINS, API_KEY, AWS_BUCKET_NAME, AWS_REGION, CDN_URL, MAX_FILE_SIZE_MB, NODE_ENV, PORT

Secret Disclosure Control

The table above lists configuration key names only. Secret values, database URLs, API keys, tokens, passwords, and private keys are intentionally not reproduced in this document.

7.2 Deployment and Runtime Configuration

Configuration Area	Evidence	Assessment
Backend Docker	backend/Dockerfile exposes port 5000 and runs compiled dist/server.js.	Found.
Frontend Docker	frontend/Dockerfile exposes port 3001 using Next standalone server.	Found.
Public Web Docker	web/Dockerfile exposes port 3000 using Next standalone server.	Found.
Filemanager Docker	filemanager/Dockerfile exposes port 3000.	Found, but production env PORT=3002 indicates port mapping must be explicit.
docker-compose	No docker-compose*.yml found.	Not Found in Current Configuration.
Backend Kubernetes Deployment	backend/k8s/deployment.yaml includes Deployment, Service, ConfigMap, Secret template, HPA and probes.	Found for backend only.
Backend Kubernetes Ingress	backend/k8s/ingress.yaml references nginx ingress and TLS secret.	Found with placeholder host api.example.com.
Azure Key Vault CSI	backend/k8s/keyvault-csi.yaml exists.	Template found; actual Key Vault values are placeholders.
CI Build	.github/workflows/ci-build.yml builds backend, frontend, web, filemanager.	Found.
Security Scan	.github/workflows/trivy-scan.yml runs weekly and PR/push Trivy scans.	Found.
Web Pipeline	web/azure-pipelines.yml and web/azure-pipelines-develop.yml.	Partial: Azure App Service/hybrid artifact and AWS ECR push indication.

7.3 Ports, Domains and TLS

Item	Value / Evidence	Notes
Public Web Port	3000	Docker EXPOSE 3000 in web/Dockerfile.
Admin CMS Port	3001	Docker EXPOSE 3001 in frontend/Dockerfile.
Backend API Port	5000	Docker EXPOSE 5000; Kubernetes Service maps port 80 to targetPort 5000.
Filemanager Port	3000 Docker EXPOSE; PORT=3002 production env	Potential runtime mapping mismatch requiring deployment clarity.
Public Domain	https://linknet.co.id	Found in frontend/public web environment.
CMS Domain	https://cms.linknet.co.id	Found in CORS/site configuration.

Item	Value / Evidence	Notes
API Domain	https://api.linknet.co.id	Found as backend base URL in frontend/web configuration.
Kubernetes Ingress Host	api.example.com placeholder	Deployment Detail Not Available for actual production ingress host.
TLS / HTTPS	Backend HTTPS redirect middleware and ingress TLS annotations found.	Certificate issuer, TLS termination point and frontend TLS config Not Found.

8. Security Architecture

Security Area	Implementation Found	Assessment
HTTPS Implementation	Backend HTTPS redirect and nginx ingress TLS/HSTS annotations.	Partial. Production TLS termination and certificate details not found.
Security Headers	Helmet and custom security headers middleware; Next.js headers found.	Found.
CORS	Production allowed origins include cms.linknet.co.id and linknet.co.id.	Found. Validate all deployed origins before go-live.
Authentication Layer	JWT access/refresh, refresh token hash/rotation, HttpOnly cookies, bcrypt password hashing, optional MFA.	Found.
Authorization	RBAC route permissions with User, Role, Permission models.	Found.
CSRF	CSRF middleware for unsafe cookie-authenticated API calls and frontend token forwarding.	Found.
Rate Limiting	Auth, upload, public form and selected public API rate limiters.	Found.
Upload Security	MIME/extension/size/magic-byte validation; ClamAV path exists but memory upload scan does not enforce ClamAV.	Partial. Add AV enforcement in production upload pipeline.
Secret Management	Environment variables and Azure Key Vault CSI template.	Partial. Actual secret manager binding and rotation policy not found.
Firewall / WAF / Network Policy	No concrete firewall/WAF/NSG/security group/network policy files found.	Not Found in Current Configuration.
S3/IAM Access	SDK integrations and bucket/CDN reference found.	IAM policy, bucket policy and block-public-access settings Not Found.
Public vs Private Access	Code defaults private storage for S3/Azure public access unless explicit opt-in.	Good application-level default; infrastructure policy still needs verification.

8.1 Authentication and Access Flow

- Admin login flow validates credentials, account status, lockout and optional MFA before issuing access and refresh tokens.

- Refresh tokens are stored as hashes and rotated, reducing impact if database token rows are exposed.
- Auth token can be supplied through HttpOnly cookie and selected bearer handling; cookie-authenticated unsafe methods require CSRF token matching.
- RBAC middleware protects CMS, dashboard, analytics, upload and administrative operations according to route-level permissions.
- OAuth or third-party identity provider integration is Not Found in Current Configuration.

9. Cloud and External Services

Service / Provider	Evidence Found	Usage / Integration	Configuration Status
AWS S3	@aws-sdk/client-s3 in backend/filemanager; filemanager bucket config.	Object upload/list/delete/presigned URL and CDN URL mapping.	Partial. Bucket exists in config; IAM/bucket policy not found.
AWS CloudFront	filemanager .env.production CDN_URL d24cmpzg3ht16e.cloudfront.net.	File delivery CDN for S3-backed assets.	Found for filemanager; primary site CDN not found.
AWS ECR	web/azure-pipelines-develop.yml references ECR registry and repository.	Container image push for public web pipeline.	Partial.
Azure Key Vault	backend services and k8s keyvault-csi.yaml template.	Secret retrieval / Kubernetes CSI secret mount pattern.	Template only; deployment values placeholders.
Azure Blob Storage	@azure/storage-blob and azureStorage.service.ts.	Optional storage backend.	Partial, enabled only when storage driver/env configured.
Google Analytics Data API	@google-analytics/data and GA4 service with GA4_PROPERTY_ID.	Analytics reporting/dashboard data.	Found.
LinkNet Enterprise API	linknetEnterprise.controller.ts.	Coverage/nearest address API through upstream OAuth token endpoint.	Found; credentials via env.
LinkNet Media API	linknetMedia.controller.ts.	Media portal data proxy with token generation.	Found; salt via env.
Yahoo Finance	web stock quote/historical API routes use yahoo-finance2.	Public stock quote and historical data proxy.	Found.
Email Service	email.util.ts logs/mock-style email behavior.	Password reset or notification utility exists.	Provider Deployment Detail Not Available.
OAuth Provider	No OAuth/social login provider implementation identified.	Not used by current auth implementation.	Not Found in Current Configuration.

10. Database and API Inventory

10.1 API Communication Surface

API / Route Group	Purpose	Security Layer
/api/v1/auth/*	Login, refresh, logout, MFA, password reset and account session flow.	Rate limit, bcrypt, JWT, refresh hash, HttpOnly cookies, CSRF-aware flow.
/api/v1/users, /roles, /permissions	User and RBAC administration.	Auth and permission middleware.
/api/v1/pages, /components, /menu, /settings	CMS page builder, public settings and menu data.	Admin auth/RBAC for mutations; public read routes where implemented.
/api/v1/news, /events, /careers, /reports, /announcements	Corporate content modules.	CMS-protected CRUD and public read flows.
/api/v1/form-modules, /contact	Dynamic forms, submissions and contact data.	Validation, upload controls, public rate limiting, CMS auth for administration.
/api/v1/upload, /filemanager, /files	Upload and file metadata operations.	Upload scanner, MIME/extension/size checks, RBAC and storage permissions.
/api/v1/dashboard, /analytics, /activity-logs	Operational dashboard, analytics and audit/activity records.	Authenticated CMS-only routes.
/health, /ready	Health and readiness endpoints.	Basic probes public; detailed diagnostics protected in production.
web/app/api/stock/*	Public website stock quote/historical proxy.	Input validation and per-IP rate limiting.
filemanager /api/*	Standalone S3 filemanager service.	x-api-key fail-closed in production plus helmet/CORS/rate limit.

10.2 Database Schema Summary

Prisma schema inventory found 57 model definitions. The full source-of-truth remains backend/prisma/schema.prisma; the table below summarizes model/table names, primary key indication, relationship evidence, and field counts.

Prisma Model	Table	Primary Key	Relationships / FK Evidence	Fields
User	users	id	Not Found in Current Configuration	32
RefreshToken	refresh_tokens	id	userId -> User.id	7
PasswordResetToken	password_reset_tokens	id	Not Found in Current Configuration	6
PasswordHistory	password_histories	id	userId -> User.id	5
Role	roles	id	Not Found in Current Configuration	10
Permission	permissions	id	Not Found in Current Configuration	8
RolePermission	role_permissions	id	permissionId -> Permission.id; roleId -> Role.id	6

Prisma Model	Table	Primary Key	Relationships / FK Evidence	Fields
UserRole	user_roles	id	roleId -> Role.id; userId -> User.id	6
Setting	settings	id	Not Found in Current Configuration	12
Menu	menus	id	parentId -> Menu?.id	24
Page	pages	id	createdById -> User.id; updatedById -> User?.id	28
PageComponent	page_components	id	pageId -> Page.id	9
Career	careers	id	Not Found in Current Configuration	18
Award	awards	id	Not Found in Current Configuration	22
ManagementCategory	management_categories	id	Not Found in Current Configuration	12
Management	managements	id	categoryId -> ManagementCategory.id	21
ContactSubmission	contact_submissions	id	userId -> User?.id	17
LogActivity	log_activities	id	userId -> User?.id	14
VisitorLog	visitor_logs	id	Not Found in Current Configuration	7
UrlRedirect	url_redirects	id	Not Found in Current Configuration	9
Folder	folders	id	parentId -> Folder?.id	12
File	files	id	createdById -> User.id; folderId -> Folder?.id	26
ContactUs	contact_us	id	Not Found in Current Configuration	16
FormModule	form_modules	id	Not Found in Current Configuration	26
FormStep	form_steps	id	formModuleId -> FormModule.id	13
FormField	form_fields	id	formModuleId -> FormModule.id; formStepId -> FormStep?.id; parentFieldId -> FormField?.id	27
FormFieldOption	form_field_options	id	fieldId -> FormField.id	12
FormFieldRule	form_field_rules	id	formModuleId -> FormModule.id; sourceFieldId -> FormField?.id; targetFieldId -> FormField?.id	14
FormResponseConfig	form_response_configs	id	formModuleId -> FormModule.id	14
FormIntegrationConfig	form_integration_configs	id	formModuleId -> FormModule.id	13
FormSubmission	form_submissions	id	formModuleId -> FormModule.id	35
FormSubmissionValue	form_submission_values	id	submissionId -> FormSubmission.id	10
FormSubmissionGroup	form_submission_groups	id	submissionId -> FormSubmission.id	10
FormSubmissionGroup Value	form_submission_group_values	id	groupId -> FormSubmissionGroup.id	10
FormSubmissionFile	form_submission_files	id	fileId -> File?.id; groupId -> FormSubmissionGroup?.id; submissionId -> FormSubmission.id	19
FormSubmissionDispatc	form_submission_dispatc	id	integrationConfigId -> FormIntegrationConfig?.id;	15

Prisma Model	Table	Primary Key	Relationships / FK Evidence	Fields
hLog	h_logs		submissionId -> FormSubmission.id	
CareerContent	career_content	id	Not Found in Current Configuration	17
ReportType	report_types	id	Not Found in Current Configuration	13
ReportSection	report_sections	id	type_id -> ReportType.id	12
news	news	id	category_id -> news_categories.id; created_by_id -> User.id; updated_by_id -> User?.id	36
news_categories	news_categories	id	Not Found in Current Configuration	13
news_highlights	news_highlights	id	news_id -> news.id	8
news_tag_relations	news_tag_relations	id	news_id -> news.id; tag_id -> news_tags.id	6
news_tags	news_tags	id	Not Found in Current Configuration	7
news_views	news_views	id	news_id -> news.id	7
events	events	id	Not Found in Current Configuration	29
event_news_relations	event_news_relations	id	event_id -> events.id; news_id -> news.id	8
event_registrations	event_registrations	id	event_id -> events.id	16
event_registration_participants	event_registration_participants	id	registration_id -> event_registrations.id	9
reports	reports	id	type_id -> ReportType?.id; section_id -> ReportSection?.id	27
AnnouncementType	announcement_types	id	Not Found in Current Configuration	14
AnnouncementSection	announcement_sections	id	type_id -> AnnouncementType.id	16
announcements	announcements	id	type_id -> AnnouncementType?.id; section_id -> AnnouncementSection?.id	19
ComponentVisibility	component_visibility	id	Not Found in Current Configuration	7
LabelGroup	label_groups	id	Not Found in Current Configuration	8
LabelNode	label_nodes	id	groupId -> LabelGroup.id; parentId -> LabelNode?.id	17
CookieConsent	cookie_consents	id	Not Found in Current Configuration	8

11. Deployment Architecture

Architecture Dimension	Assessment	Evidence
Monolith vs Microservice	Service-separated multi-application architecture, not a single monolith.	Separate backend, frontend, web, and filemanager packages with individual Dockerfiles.
Frontend / Backend Separation	Separated.	Admin CMS and public website call backend API via configured

Architecture Dimension	Assessment	Evidence
		base URLs.
Containerization	Present.	Dockerfiles for all four components.
Orchestration	Partial.	Backend Kubernetes manifests only. No complete cluster-level topology or frontend/filemanager manifests.
Shared Hosting / VPS / Cloud	Deployment Detail Not Available.	Wamp path indicates local workspace; production provider not conclusively defined.
Cloud Deployment Indication	Partial AWS and Azure indications.	AWS S3/CloudFront/ECR, Azure Pipeline/App Service artifact, Azure Key Vault/Blob references.
Scalability	Backend horizontal scaling evidenced; full system scalability partially defined.	Backend HPA min 3 max 10; frontend/filemanager autoscaling not found; DB/storage HA not found.
High Availability	Partial.	Backend replicas 3; load balancer, multi-AZ database, cache HA and storage policy not found.
Observability	Partial.	Health endpoints, logs, activity logs and GA4 analytics found. Centralized metrics/APM not found.

11.1 CI/CD and Security Pipeline

Pipeline	Evidence	Notes
GitHub CI Build	.github/workflows/ci-build.yml	Build matrix includes backend, frontend, web and filemanager.
Trivy Security Scan	.github/workflows/trivy-scan.yml	Scans dependency vulnerabilities and secrets on push/PR and weekly schedule.
Azure Pipeline	web/azure-pipelines.yml	Next.js SSR/hybrid artifact for Azure App Service/platform path.
AWS ECR Pipeline	web/azure-pipelines-develop.yml	Build, SAST/Trivy and Docker push to AWS ECR registry in ap-southeast-3.
Deployment Approval / Rollback	Not found.	Release gates, rollback runbooks and promotion controls Not Found in Current Configuration.

12. Infrastructure Risks and Recommendations

Risk Framing

Risks below are infrastructure and deployment risks, not confirmed exploitation findings. Severity reflects potential audit/security impact if the missing deployment controls are not implemented or evidenced.

Risk	Severity	Evidence / Concern	Recommendation
Incomplete edge topology evidence	Medium	DNS provider, production load balancer, public IPs and frontend ingress are not defined in repo.	Document actual DNS records, CDN/LB, ingress, TLS termination, and routing paths in infrastructure-as-code or a controlled runbook.
Firewall/WAF/network policy not evidenced	Medium	No WAF, firewall, NSG/security group or Kubernetes NetworkPolicy definitions found.	Add WAF and network policy evidence; restrict backend/database/storage to required sources only.
Database topology unavailable	Medium	DATABASE_URL exists, but host, SSL mode, subnet, HA, backup and retention are not visible.	Define DB architecture, private access, SSL requirement, backup/restore, HA and monitoring controls.
Secrets and env governance risk	High	Environment files exist in workspace and contain deployment-sensitive keys; values are not reproduced here.	Move production secrets to managed secret store; ensure .env production files are excluded from Git and rotated if ever exposed.
S3 IAM and bucket policy unavailable	Medium	S3 integration and bucket/CDN reference found, but IAM/bucket policy is not present.	Attach least-privilege IAM policy, block public access unless intentionally public through CDN, and log S3 access.
Filemanager port mismatch	Medium	Docker EXPOSE 3000 but .env.production PORT=3002.	Clarify container port mapping and service targetPort; align Dockerfile, runtime env and ingress/service definition.
Backend ingress uses placeholder host	Medium	backend/k8s/ingress.yaml references api.example.com.	Replace placeholder with actual controlled host and TLS secret reference; verify HSTS and TLS policy at edge.
Frontend/filemanager orchestration missing	Medium	Kubernetes manifests exist only for backend.	Add deployment manifests or platform specs for frontend, public web and filemanager, including resource limits, probes, scaling and network rules.
Queue deployment unclear	Low	Bull/ioredis optional paths found, but Redis deployment not defined.	If REDIS_ENABLED=true in production, document Redis topology, authentication, persistence, HA and network restrictions.
Upload malware scanning partial	Medium	MIME/magic validation exists; ClamAV path exists but memory-upload route scanning does not fully enforce AV.	Enforce AV scanning/quarantine for all accepted upload flows before public access or downstream processing.
Single point of failure not fully assessed	Medium	Backend HPA exists, but DB/cache/storage edge HA details are unavailable.	Add HA design for DB, cache, file service, CDN/LB and backup/restore objectives.

12.1 Hardening Recommendations

- Create infrastructure-as-code for edge, DNS, load balancer, TLS certificates, firewall/WAF, Kubernetes ingress and network policies.
- Use a managed secrets platform as the only production secret source and rotate secrets that may have existed in local or historical deployment files.
- Restrict database access to backend runtime identities/subnets and require TLS for database connections where supported.
- Apply least-privilege IAM to S3 and CloudFront, enforce S3 block-public-access, and use CDN/origin access controls for public assets.
- Publish a deployment runbook covering domains, ports, health checks, rollback, backup/restore, scaling thresholds and incident escalation.
- Add monitoring/APM/log aggregation for backend, frontend, filemanager, database, storage and external API failures.
- Standardize all component deployments with health probes, resource requests/limits, autoscaling rules and private service discovery.

13. Appendix

13.1 Evidence Files Reviewed

Area	Files / Paths
Backend Source	backend/src/server.ts, backend/src/routes, backend/src/controllers, backend/src/services, backend/src/middleware
Database	backend/prisma/schema.prisma and migrations
Frontend	frontend/package.json, frontend/Dockerfile, frontend environment files and service/config code
Public Web	web/package.json, web/Dockerfile, web/app API routes, web Azure Pipeline files
Filemanager	filemanager/package.json, filemanager/Dockerfile, filemanager/src and environment files
Containers	backend/Dockerfile, frontend/Dockerfile, web/Dockerfile, filemanager/Dockerfile
Kubernetes	backend/k8s/deployment.yaml, backend/k8s/ingress.yaml, backend/k8s/keyvault-csi.yaml
CI/CD	.github/workflows/ci-build.yml, .github/workflows/trivy-scan.yml, web/azure-pipelines*.yml

13.2 Glossary

Term	Definition
CDN	Content Delivery Network used to cache/distribute static or media assets closer to users.
Ingress	Kubernetes edge routing resource that exposes services through HTTP/HTTPS host/path routing.

Term	Definition
RBAC	Role-Based Access Control; application permissions mapped through roles and user assignments.
CSRF	Cross-Site Request Forgery; mitigated here through CSRF token validation for unsafe cookie-authenticated requests.
Prisma	Node.js ORM used by the backend to access PostgreSQL.
HPA	Horizontal Pod Autoscaler; Kubernetes scaling resource found in backend deployment manifest.
S3	AWS object storage service used by backend/filemanager integration paths.
Key Vault CSI	Kubernetes CSI driver pattern for mounting Azure Key Vault secrets into pods.

13.3 Explicitly Unavailable Details

- Production DNS provider, authoritative zone configuration, public IPs and DNS record TTLs.
- Production CDN/load balancer configuration for main public website and CMS.
- Firewall, WAF, security group, network security group, Kubernetes NetworkPolicy or egress firewall definitions.
- Production database host topology, subnet, SSL enforcement, backup/restore policy and HA setup.
- Actual IAM policies for AWS S3/CloudFront/ECR and workload identity bindings.
- Actual Azure Key Vault instance name, tenant/workload identity binding and secret rotation process.
- Complete deployment manifests for frontend, public web and filemanager runtimes.